

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q12-0093

Boonsboro 138 kV
November 16, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0093. The Interconnection Customer proposes a 100.3 MW wind facility interconnecting at the Boonsboro 138 kV substation in the Virginia area. The request was received on January 11, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 67 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$7,575,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

BEAR SUBSTATION to LEAVITT 345 KV #1: 114.8% loading (1,580 MVA) following the outage of the BEAR SUBSTATION to LEAVITT SUBSTATION 345 KV Line 2; DFAX = 14.8%.

NORTH ANNA to LOUISA GENERATION FACILITY 345 KV: 104.4% loading (1,686 MVA) following the outage of the CARRABELLE to LOUISA GENERATION FACILITY 345 KV; DFAX = 13.6%.

The CARRABELLE to LOUISA GENERATION FACILITY 345 KV is overloaded to 104.0% of its 1,492 MVA emergency rating for the loss of the LOUISA GENERATION FACILITY to N ANNA 345 KV. The project contributes 11.5% DFAX (11.5 MW) to this overload.

The PERSON to MECKLENBURG POWER STA 345 KV is overloaded to 102.5% of its 2,318 MVA emergency rating for the loss of the BEAR to CLOVER 345 KV. The project contributes 23.3% DFAX (23.4 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
BEAR SUBSTATION to LEAVITT 345 KV #1	Rebuild line with higher-capacity conductor	\$208,732,000	2% (\$5,265,000)
NORTH ANNA to LOUISA GENERATION FACILITY 345 KV	Reconductor line	\$32,424,000	2% (\$681,000)
LOUISA GENERATION FACILITY to CARRABELLE 345 KV	Reconductor line	\$38,236,000	2% (\$798,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
PERSON to MECKLENBURG POWER STA 345 KV	Reconductor line	\$23,469,000	4% (\$831,000)
	Total allocated to project		\$7,575,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.